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Amsterdam Business School

Picture source: Unsplash / James Baltz

MScBA: Ethics and the Future of Business

SESSION 4: **CHALLENGES & SOLUTIONS**

Arno Kourula – 7 September 2026





Recap of last week

1. Changing behavior

2. Corporate responsibility and sustainability

- Drivers of corporate responsibility
- Corporate responsibility and digital economy

3. Stakeholder thinking

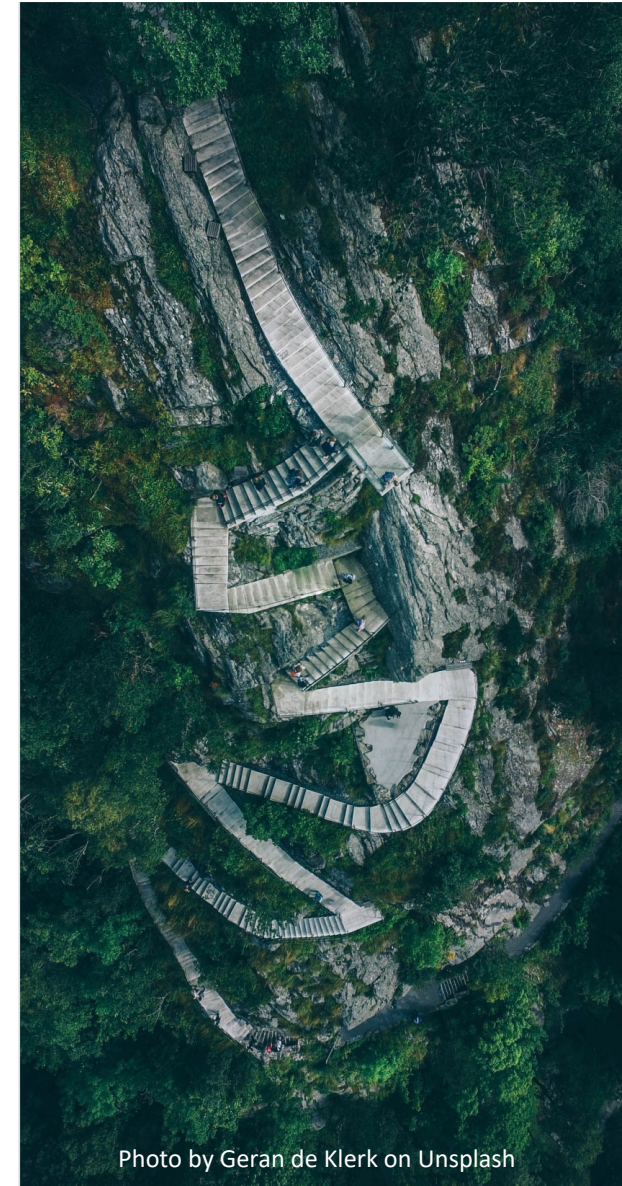


Photo by Geran de Klerk on Unsplash



Course practicalities

1. General part exam

- Exam will be in person (see rooster.uva.nl) using Ans software.
- It will be Wednesday, September 23, 19:00-21:00, Testamenzaal USC Sporthal 1&2, Science Park.
- Example exam questions available in Canvas (see Modules).
- Reflection questions for each session on Canvas (see Modules).
- Resit in January 2027 (see rooster.uva.nl).

2. What's next after the general lectures and exam?

- Track-specific parts, to be updated in Modules.
- See rooster for schedule, end of September / early October.



Photo source: abs.uva.nl

Agenda for today

1. Broader context of business

- Anthropocene and key sustainability trends
- Frameworks for sustainability

2. Business and climate change

- Climate change science
- Climate change targets

3. Towards a sustainable economy?

- Circular economy
- Sustainable business models
- Systems change

4. Recap and closing of general part



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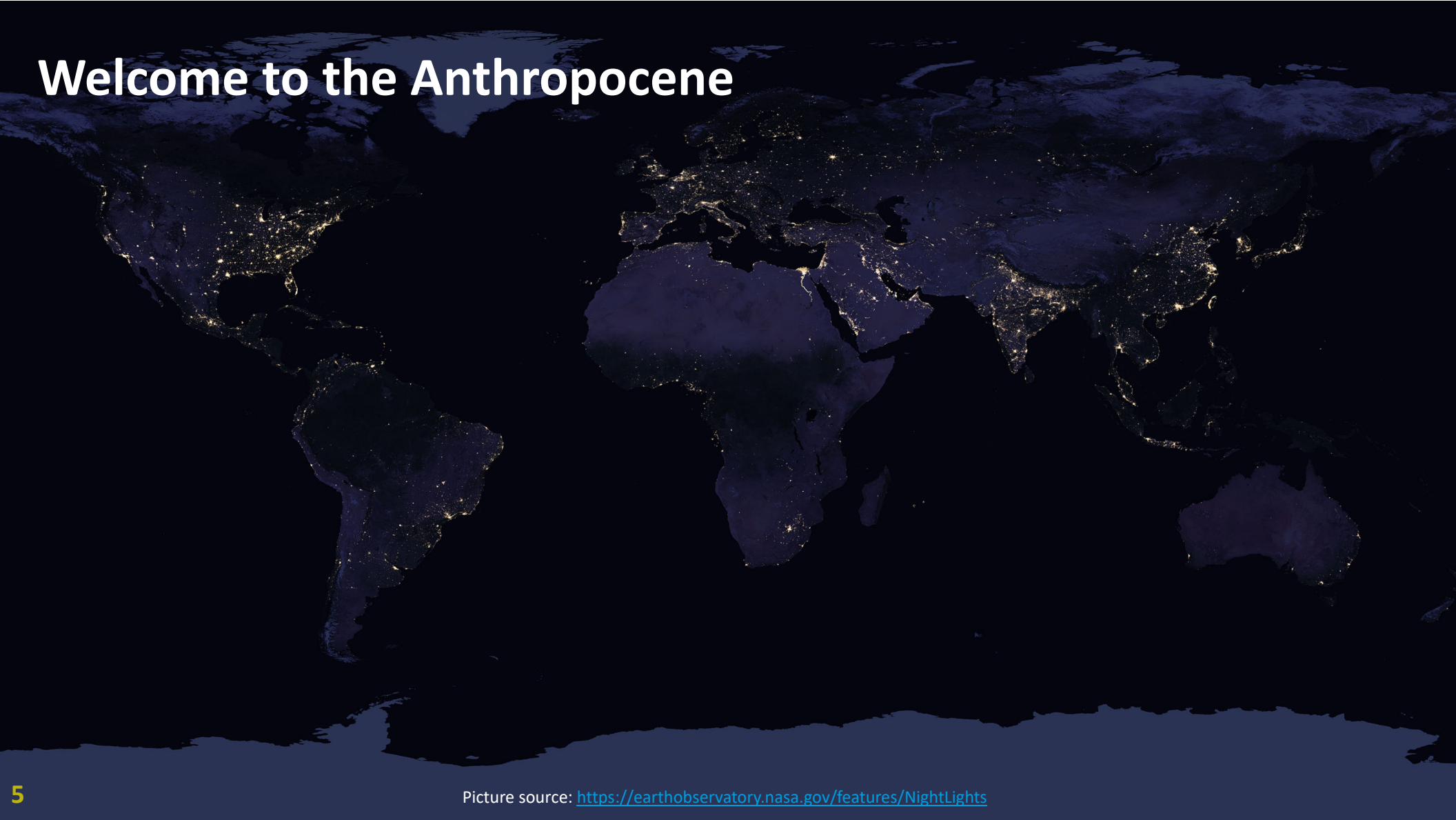
1. Broader context of business

Picture source:

<https://www.theguardian.com/environment/2016/jan/07/human-impact-has-pushed-earth-into-the-anthropocene-scientists-say>



Welcome to the Anthropocene



Picture source: <https://earthobservatory.nasa.gov/features/NightLights>

Welcome to the Anthropocene

A new epoch of geological time (epoch = tens of millions of years).

Examples of human impact:

- 432 ppm of CO₂ in atmosphere (highest in at least 2 million years).
- Our global economy consumes 100+ billion tons of materials a year and wastes over 90%. All human-made mass exceeds all living biomass (plants, bacteria, animals, etc.).
- Agriculture, deforestation, mining and urbanization: Less than 3% of land on earth is ecologically intact. Rapid ecosystem deterioration.
- Rapid species extinction (100 to 1000 times faster): humans have wiped out an average of 68% of (monitored) mammals, birds, fish and amphibians since 1970. Mammal biomass on planet: 62% livestock (+pets), 34% humans, 4% wild mammals.
- Traces of estrogen, ibuprofen and caffeine in fish.

“If all insects on Earth disappeared, within 50 years all life on Earth would end. If all human beings disappeared from the Earth, within 50 years all forms of life would flourish.” Biologist Jonas Salk



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Hockey sticks!

nature

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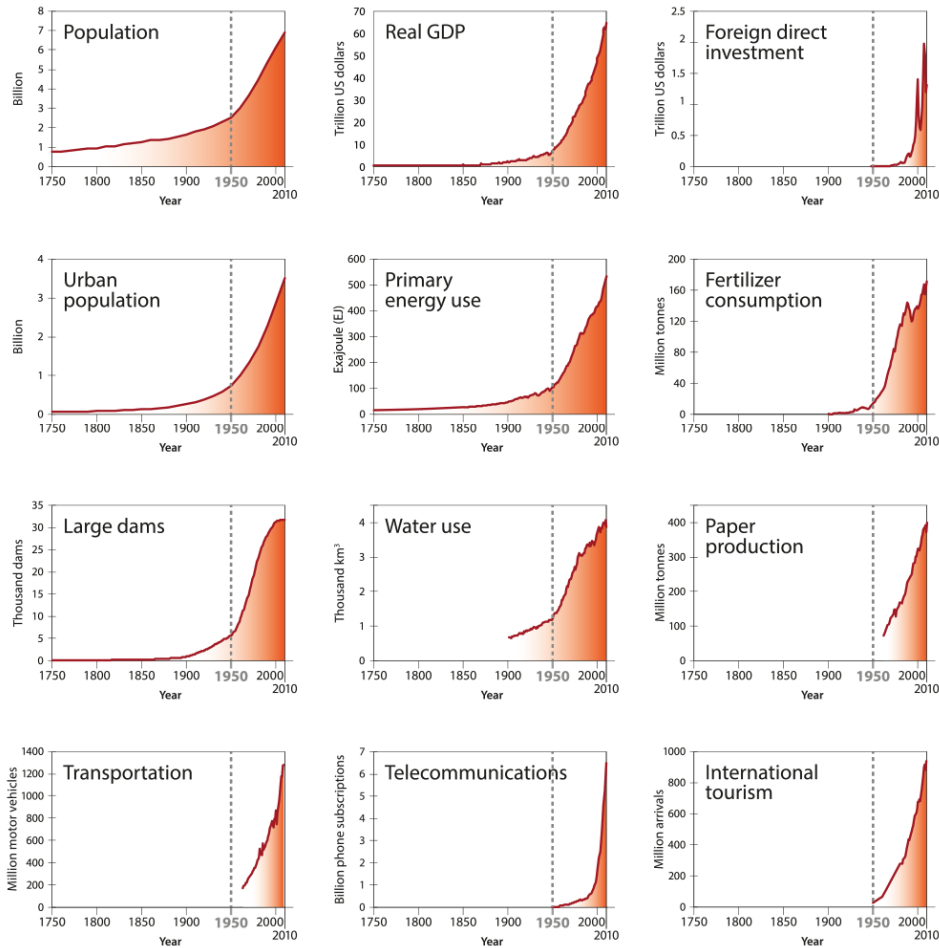
NEWS | 06 March 2024 | Correction [06 March 2024](#)

Geologists reject the Anthropocene as Earth's new epoch – after 15 years of debate

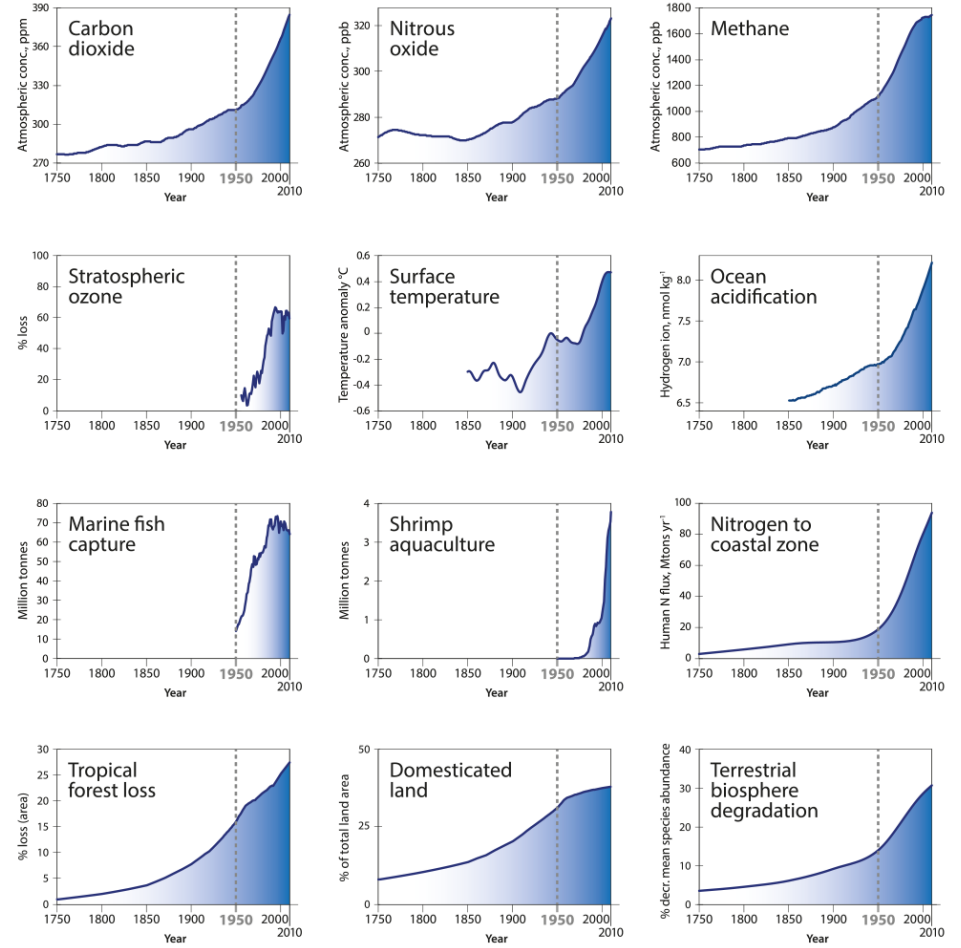




Socio-economic trends

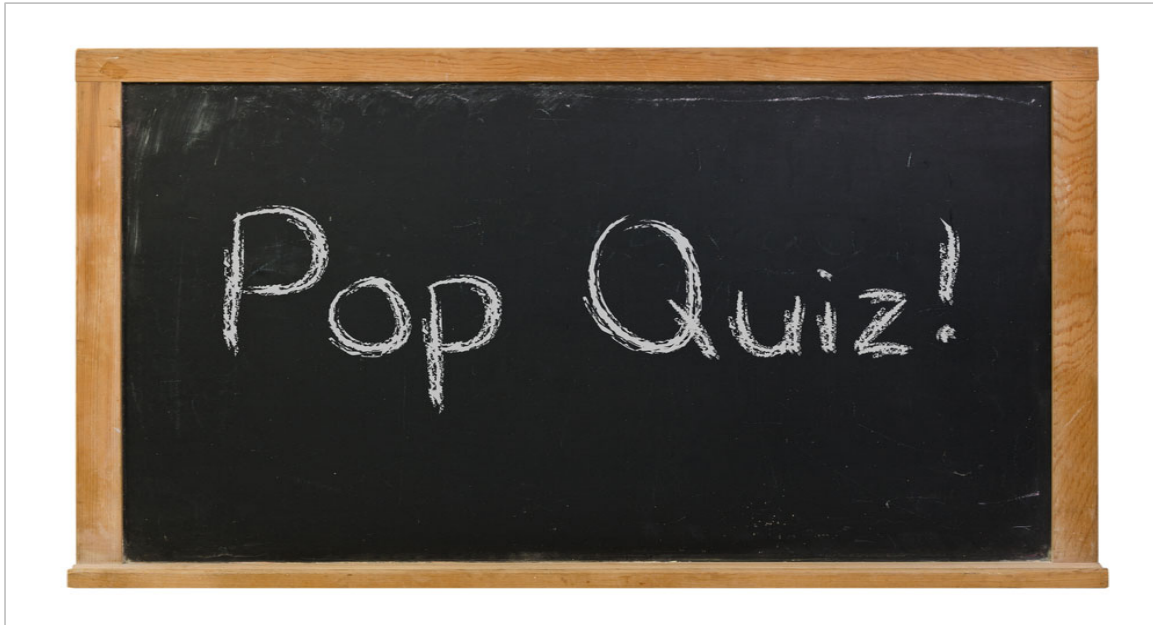


Earth system trends



Updated Great Acceleration Graphs

Source: Will Steffen et al. "The trajectory of the Anthropocene: The Great Acceleration." The Anthropocene Review, March 2015



Worldview upgrader: <https://upgrader.gapminder.org>

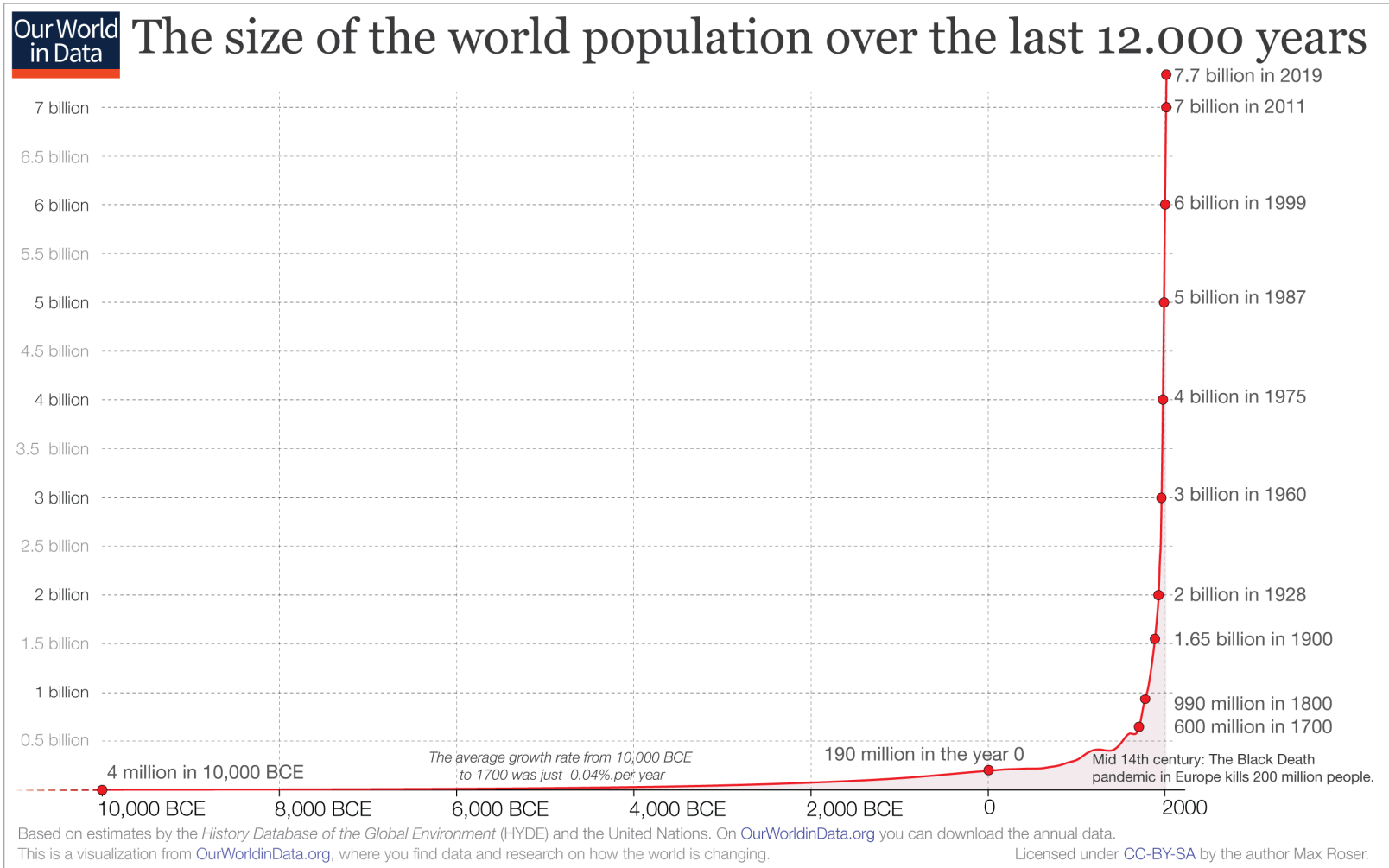


In the past century...

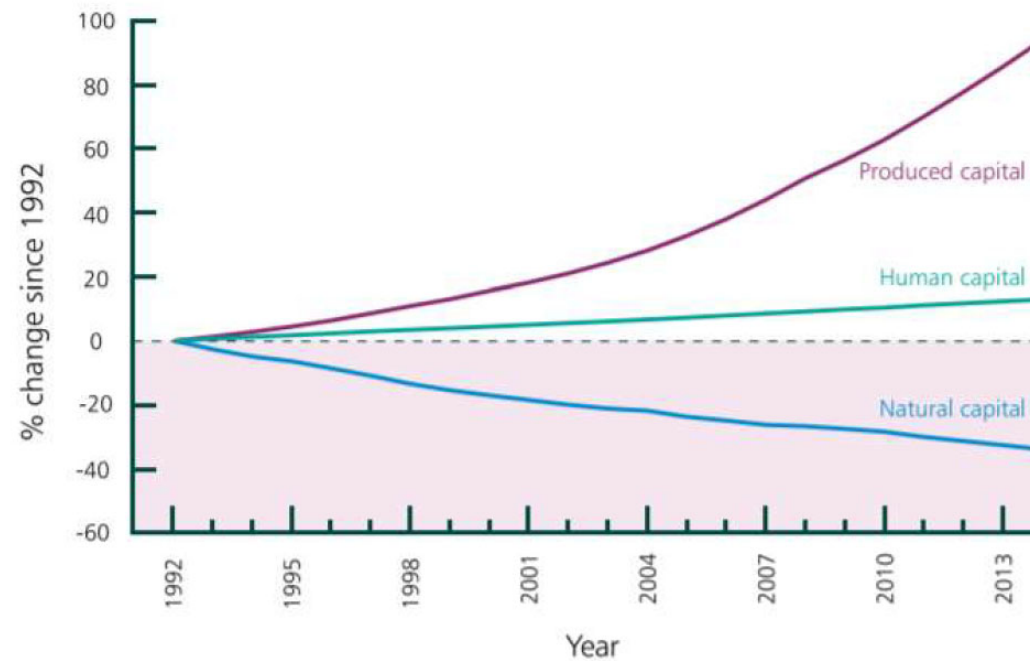
- World population more than quadrupled.
- World economy increased by a factor of 80.
- Average life expectancy more than doubled.
- The share of people living in extreme poverty dropped by a factor of 8.
- Literacy rates more than quadrupled.
- Astonishing developments in science and technology.

Source: https://en.wikipedia.org/wiki/Amsterdam_Centraal_station

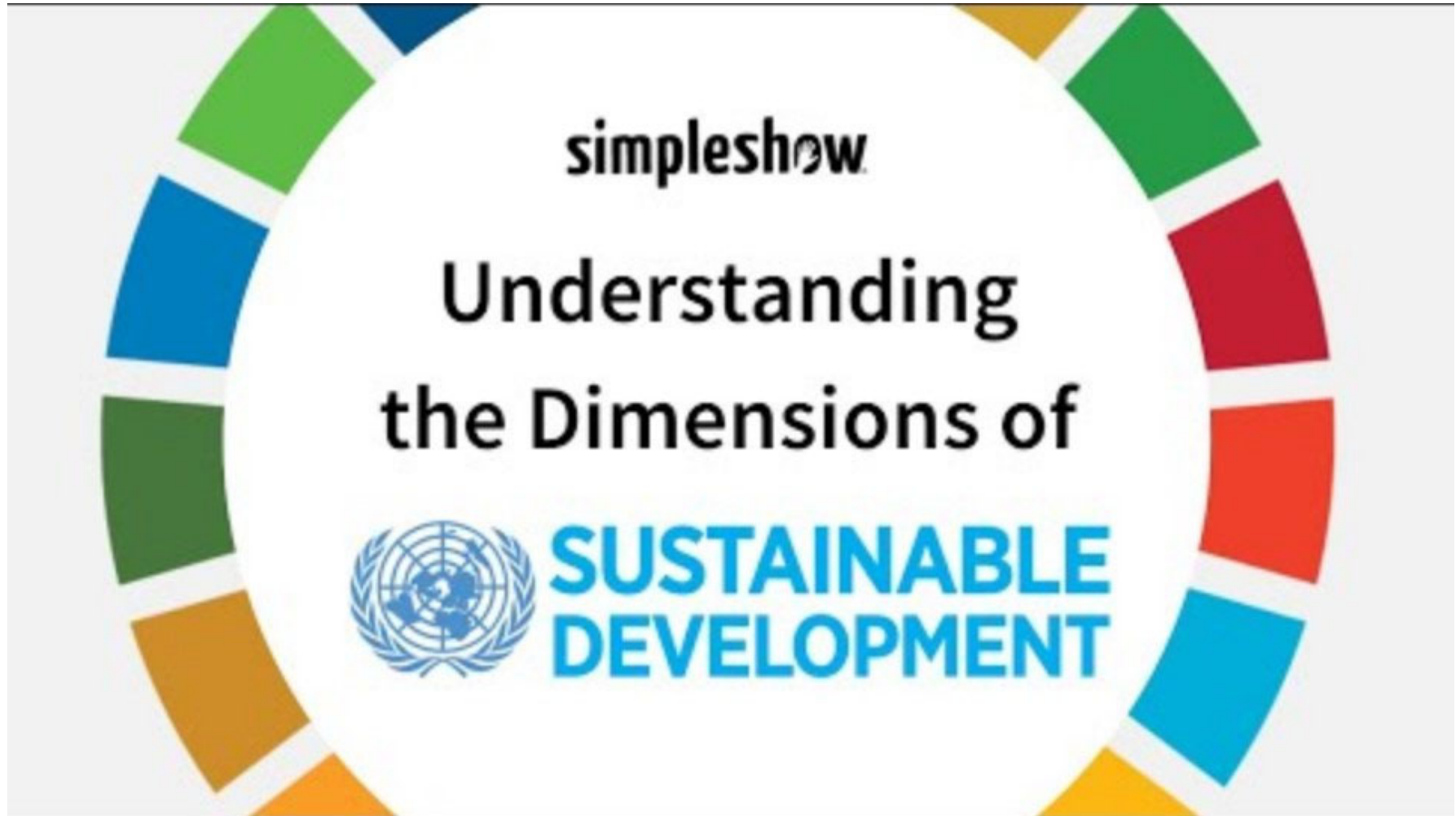
Population growth:



Global wealth per capita



Global Wealth per Capita (Managi & Kumar, 2018)

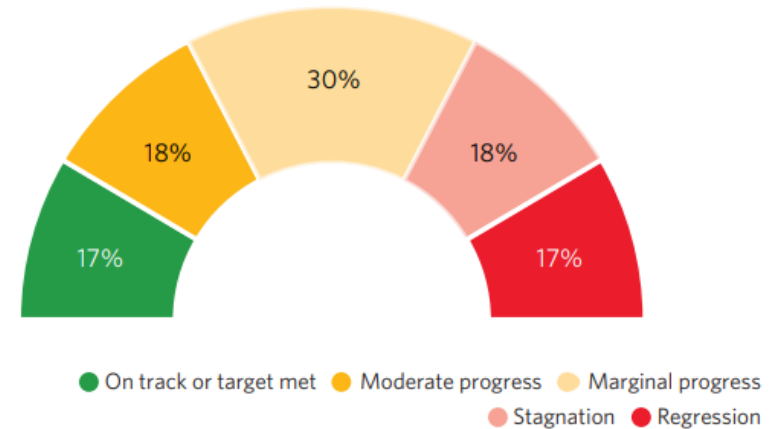


Introduction: www.youtube.com/watch?v=pgNLonYOc9s/

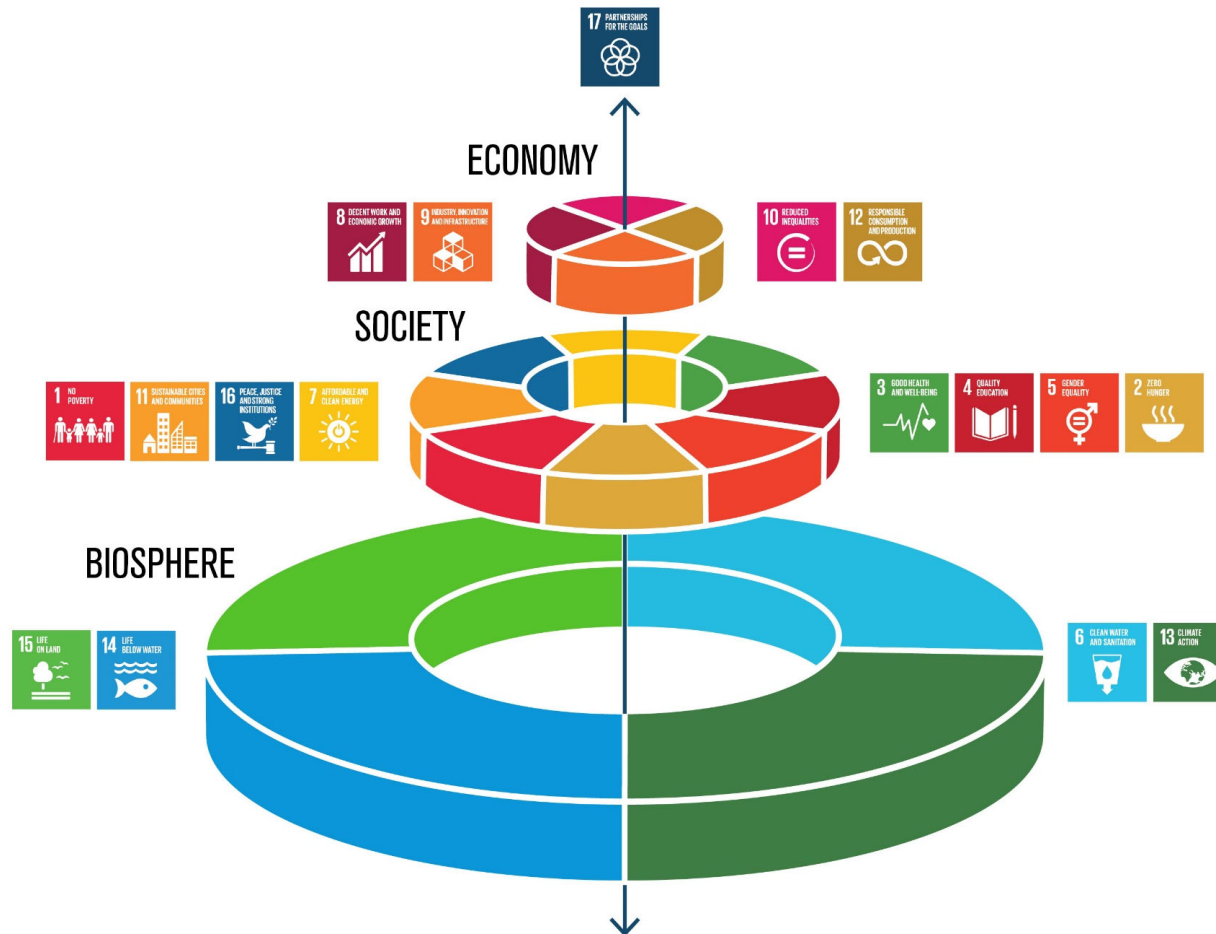
United Nations Sustainable Development Goals (SDGs)



Overall progress across targets based on 2015–2024 global aggregate data

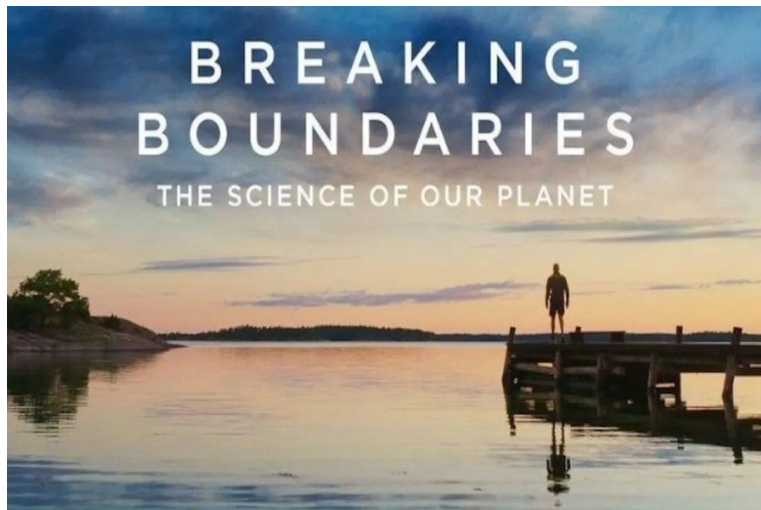


SDG wedding cake

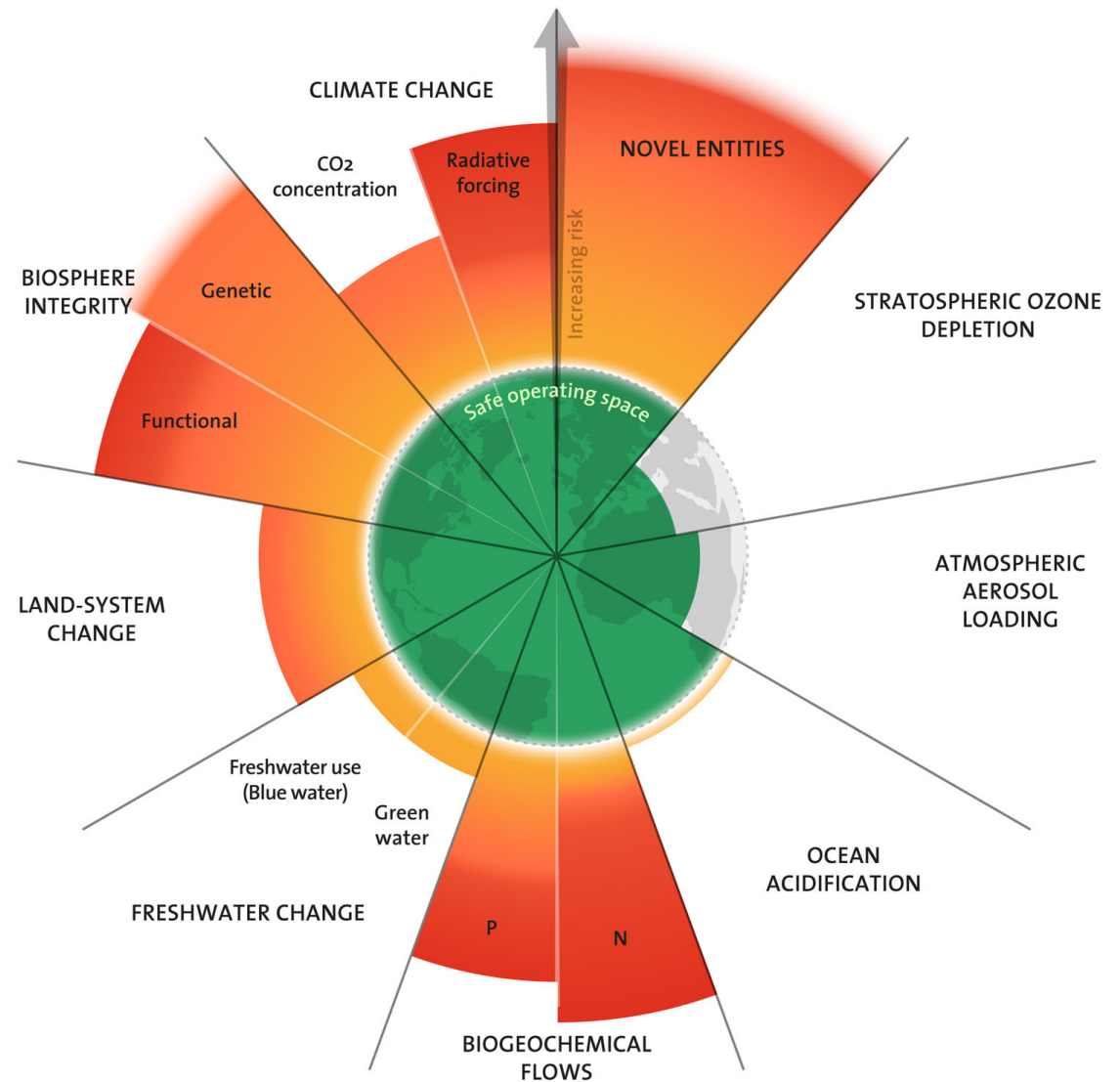


Source: www.stockholmresilience.org/research/research-news/2016-06-14-how-food-connects-all-the-sdgs.html

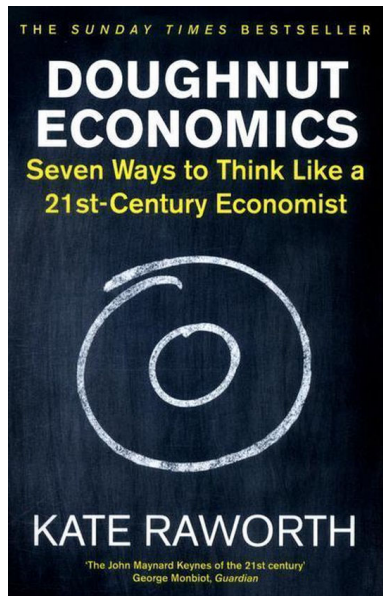
Planetary boundaries: 7 breached



Sources: www.imdb.com/title/tt14539726/
Azote for Stockholm Resilience Centre, based on analysis in
Sakschewski and Caesar et al. 2025

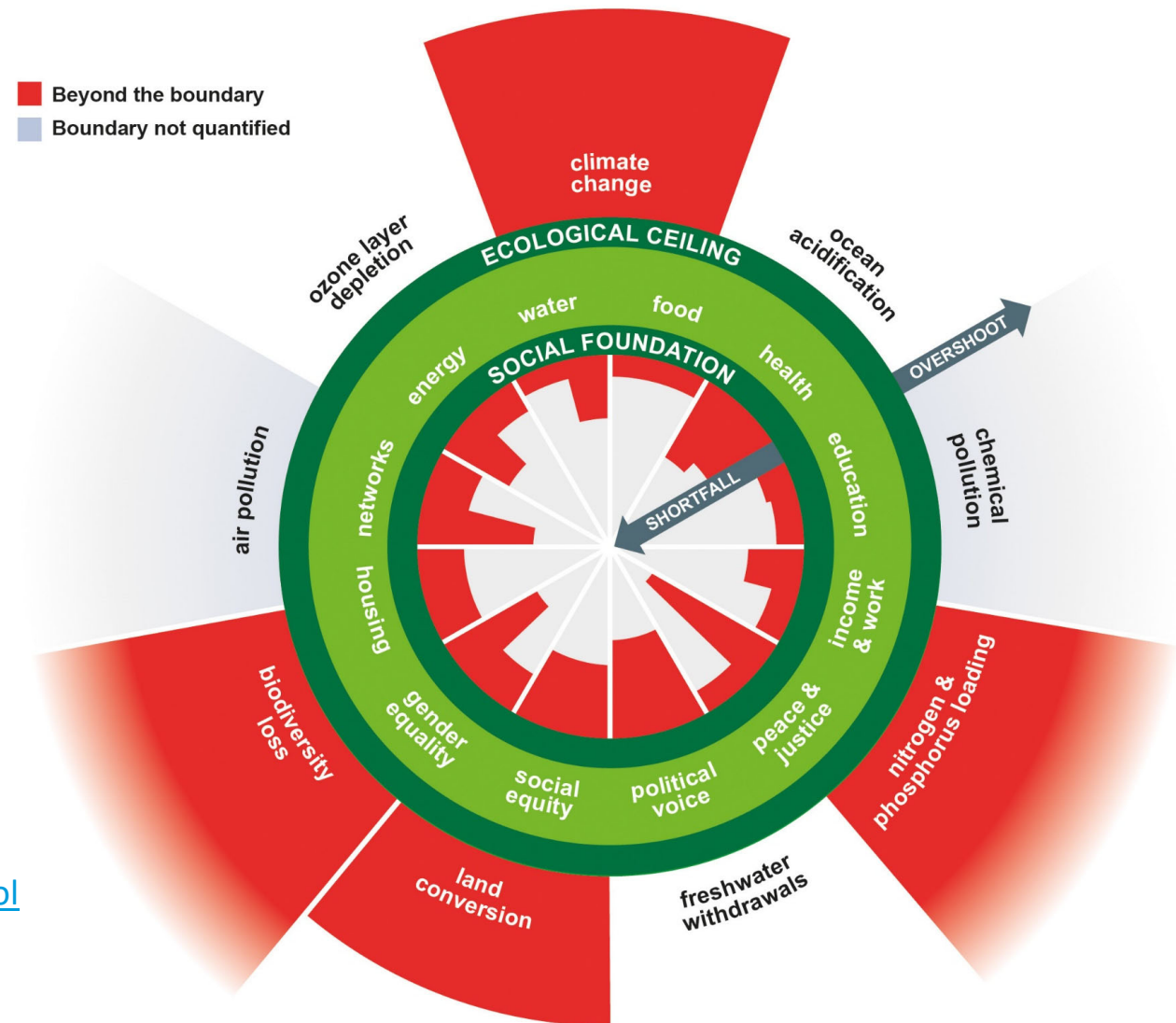


Doughnut Economics

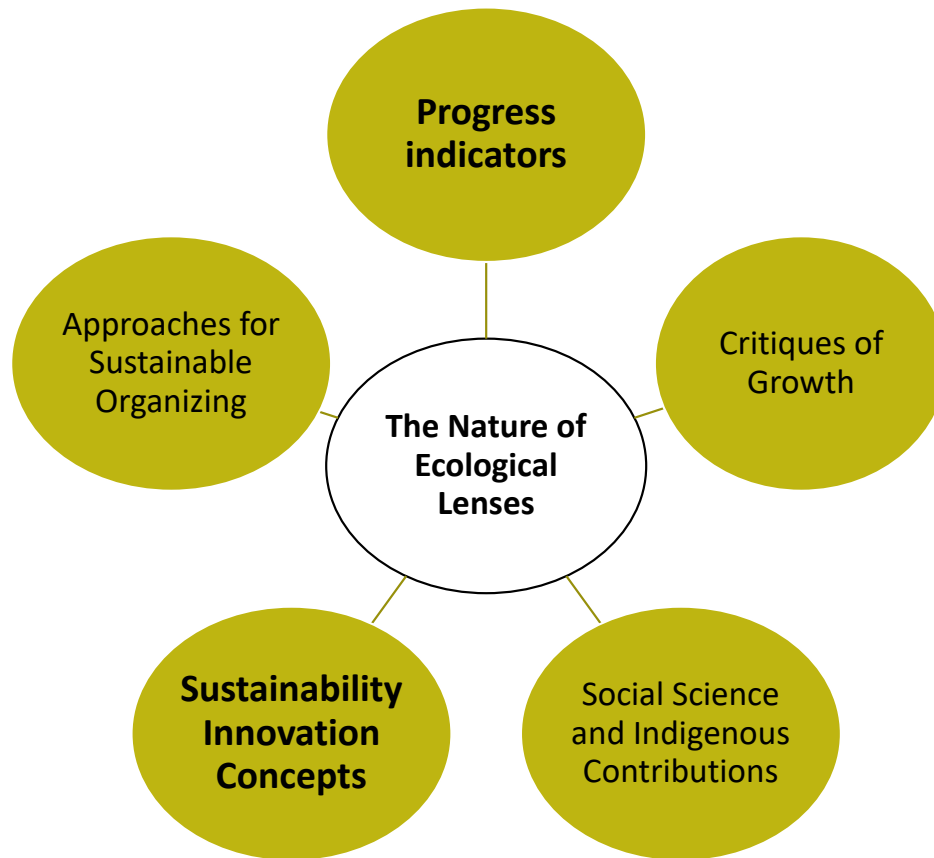


Check your country's doughnut:

<https://doughnuteconomics.org/tools-and-stories/22>



Lenses in (Corporate) Sustainability



- A range of ecological paradigms, theories, concepts and approaches have been proposed to examine the role of corporations in environmental sustainability which are here referred to as *lenses*.
- Twelve lenses, especially prevalent in the field of management, have been selected and categorized according to their nature.



Reflection question

What is the best **organizing frame(work)** for corporate sustainability activities? Is it the Corporate Responsibility, Corporate Sustainability, ESG, SDGs, Planetary Boundaries, Doughnut Economics, or some other lens?



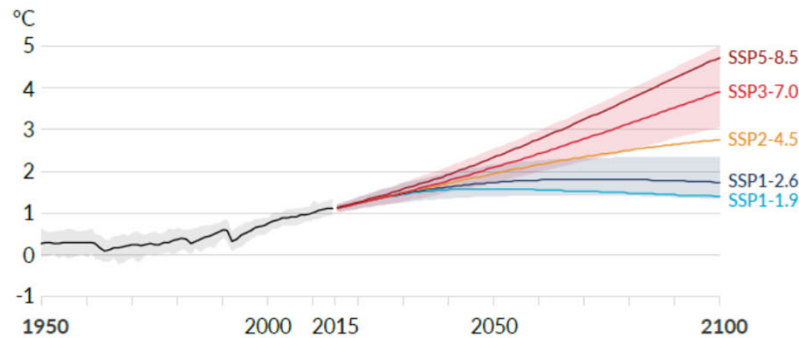
2. Business and climate change

Picture source: <https://todaysenvironmentalist.com/2018/11/07/wwf-report-reveals-staggering-extent-of-human-impact-on-planet/>

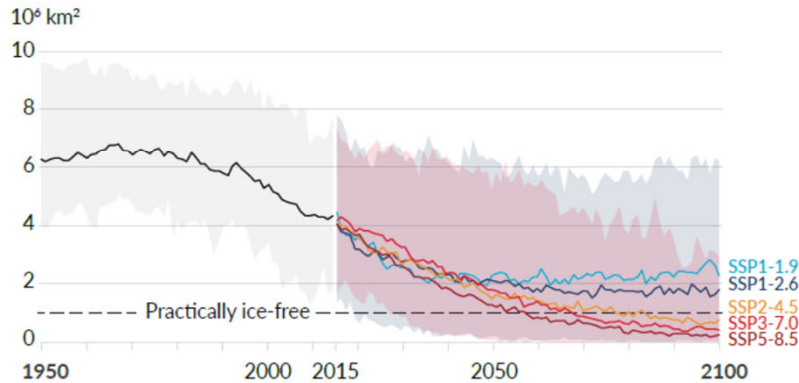


Climate Change: Key trends

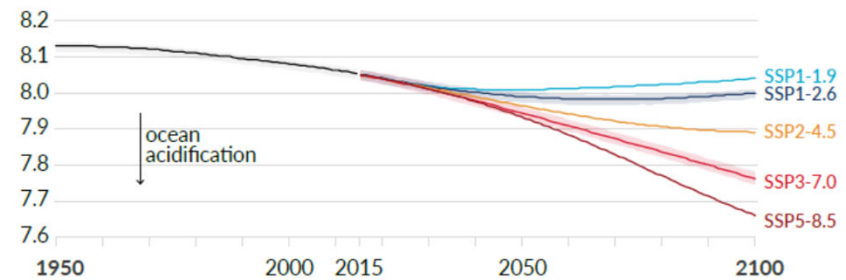
a) Global surface temperature change relative to 1850-1900



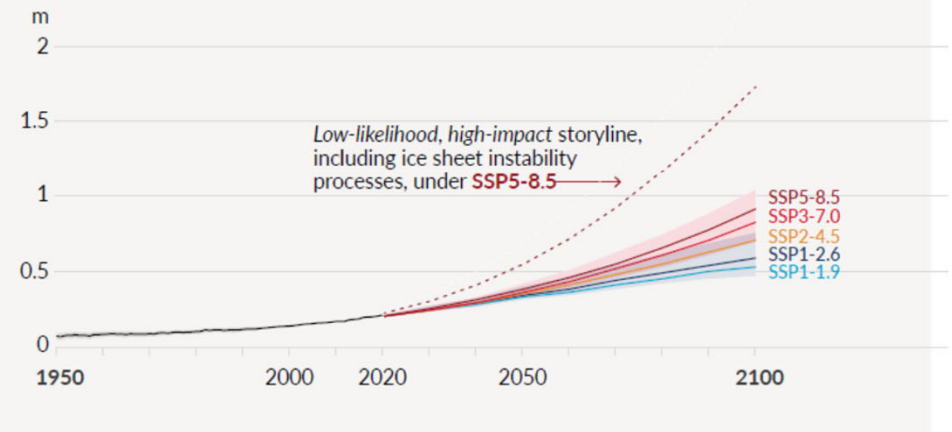
b) September Arctic sea ice area



c) Global ocean surface pH (a measure of acidity)



d) Global mean sea level change relative to 1900

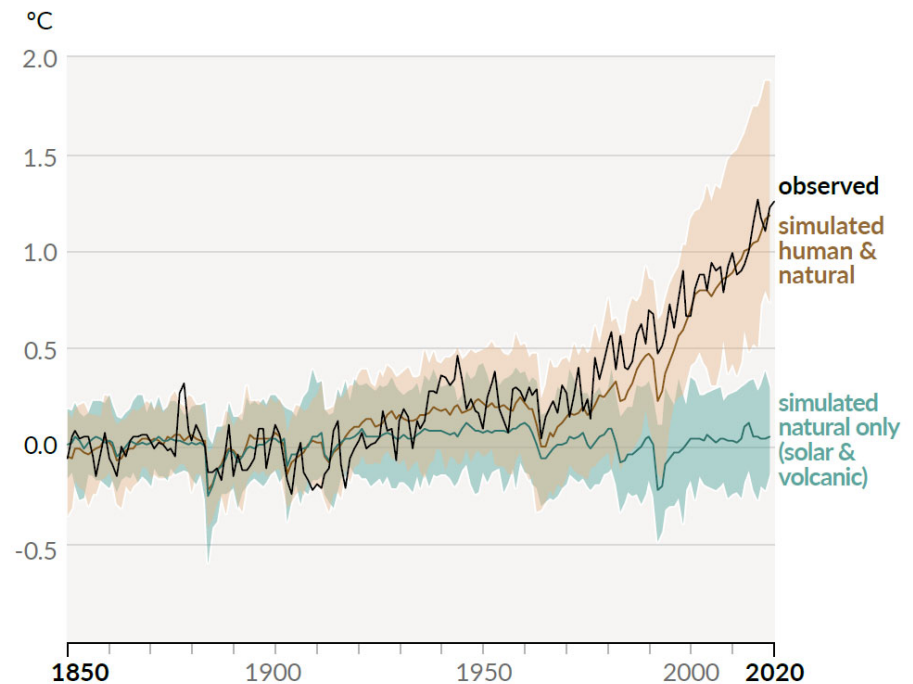


Source: IPCC. 2021. Climate change 2021 – The physical science basis: summary for policymakers. Intergovernmental Panel on Climate Change.

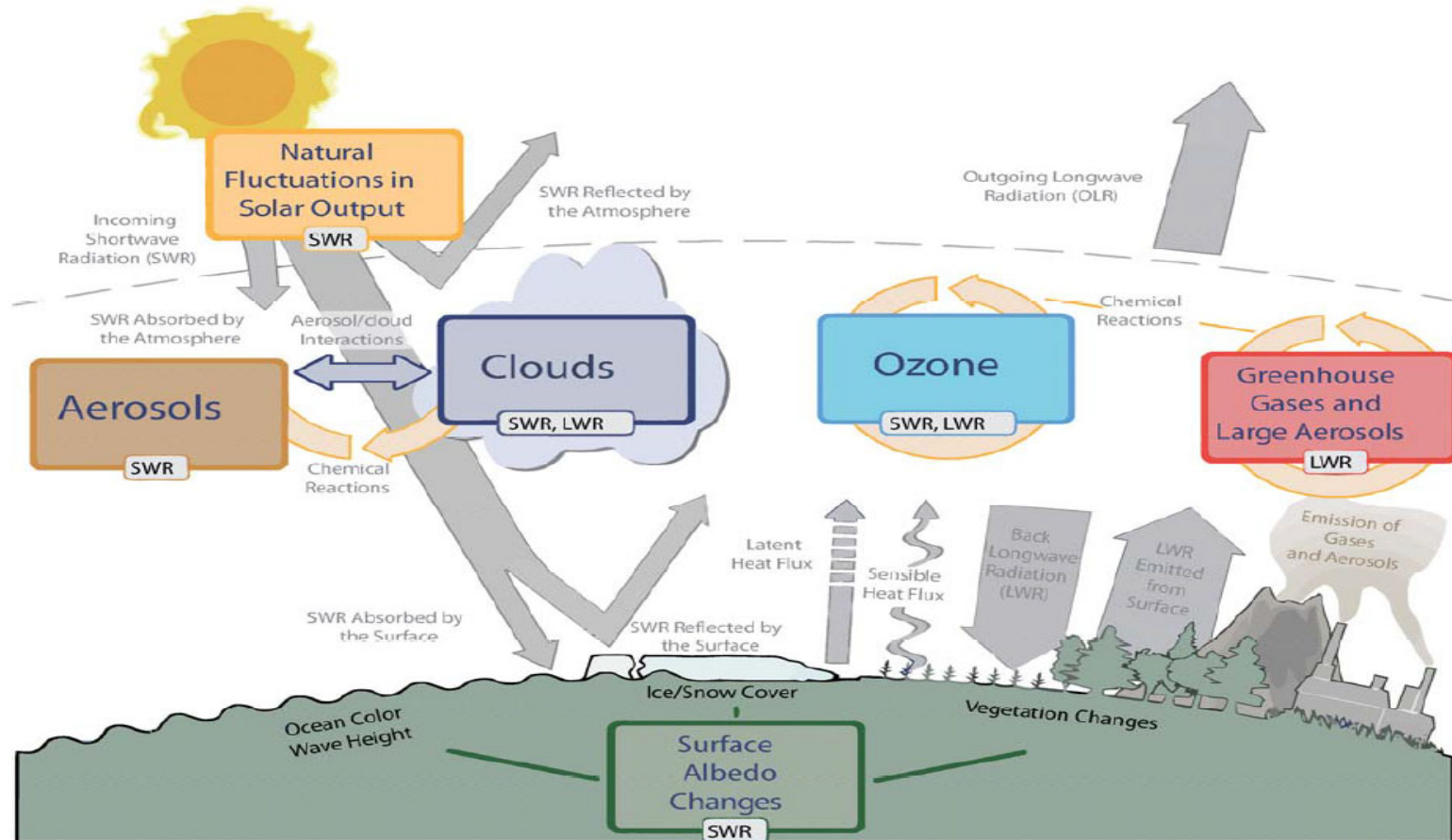
Observed changes and their causes (IPCC 2023)

“It is unequivocal that **human influence** has warmed the atmosphere, ocean and land. **Widespread and rapid changes** in the atmosphere, ocean, cryosphere and biosphere have occurred.”

b) Change in global surface temperature (annual average) as **observed** and simulated using **human & natural** and **only natural** factors (both 1850-2020)



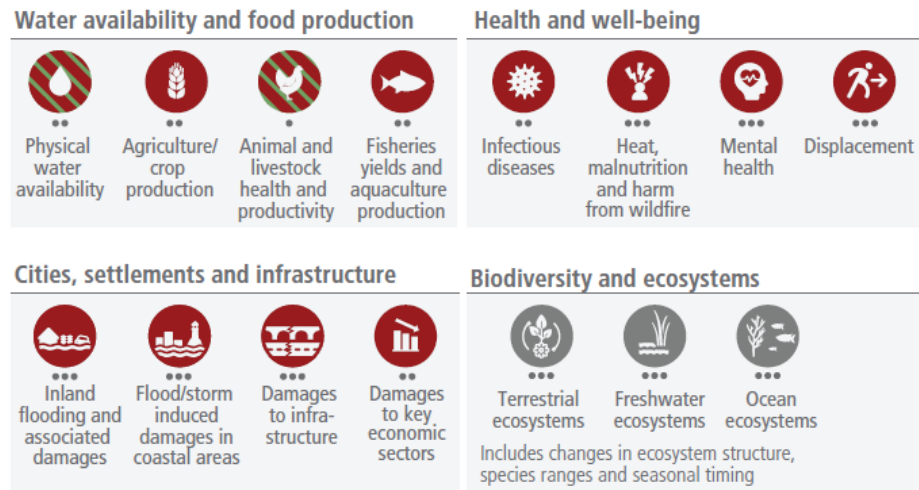
Greenhouse effect (See also video: www.youtube.com/watch?v=BPJIM_hCFj0)



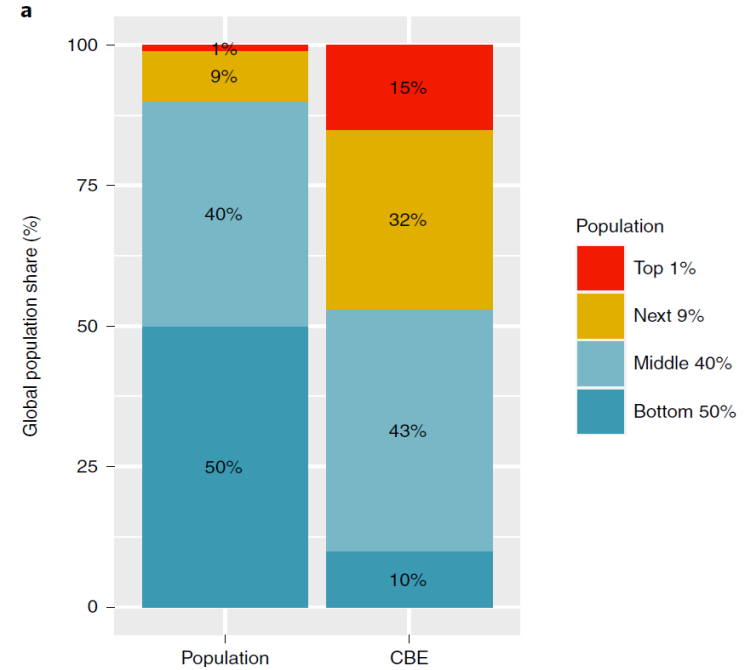
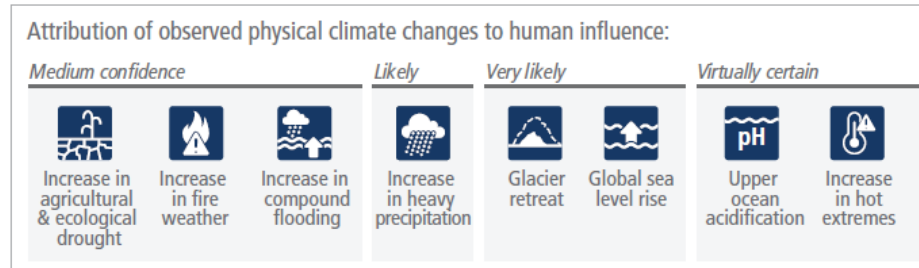
IPCC (2013), see https://www.ipcc.ch/site/assets/uploads/2018/02/WG1AR5_all_final.pdf

Impacts of climate change

a) Observed widespread and substantial impacts and related losses and damages attributed to climate change

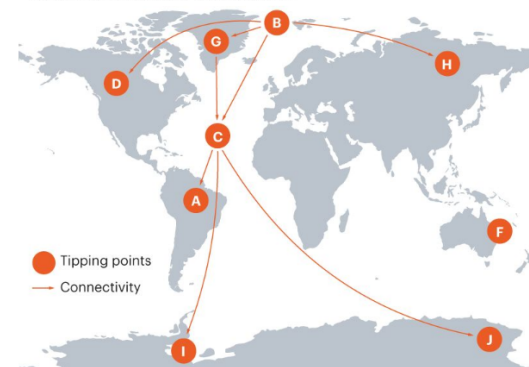


b) Impacts are driven by changes in multiple physical climate conditions, which are increasingly attributed to human influence



RAISING THE ALARM

Evidence that tipping points are under way has mounted in the past decade. Domino effects have also been proposed.



Climate-Related Financial Disclosures

Core Elements of Recommended Climate-Related Financial Disclosures



Governance

The organization's governance around climate-related risks and opportunities

Strategy

The actual and potential impacts of climate-related risks and opportunities on the organization's businesses, strategy, and financial planning

Risk Management

The processes used by the organization to identify, assess, and manage climate-related risks

Metrics and Targets

The metrics and targets used to assess and manage relevant climate-related risks and opportunities

TCFD 2016

Scope 1: direct emissions from owned or controlled sources / Scope 2: indirect emissions from the purchase and use of electricity, steam, heating and cooling / Scope 3: all other indirect emissions that occur in the upstream and downstream activities.



3. Towards a sustainable economy?

Picture source: <https://www.tvo.org/article/what-happens-when-humans-rule-the-earth-documenting-the-anthropocene>



Linear Economy



Recycling Economy

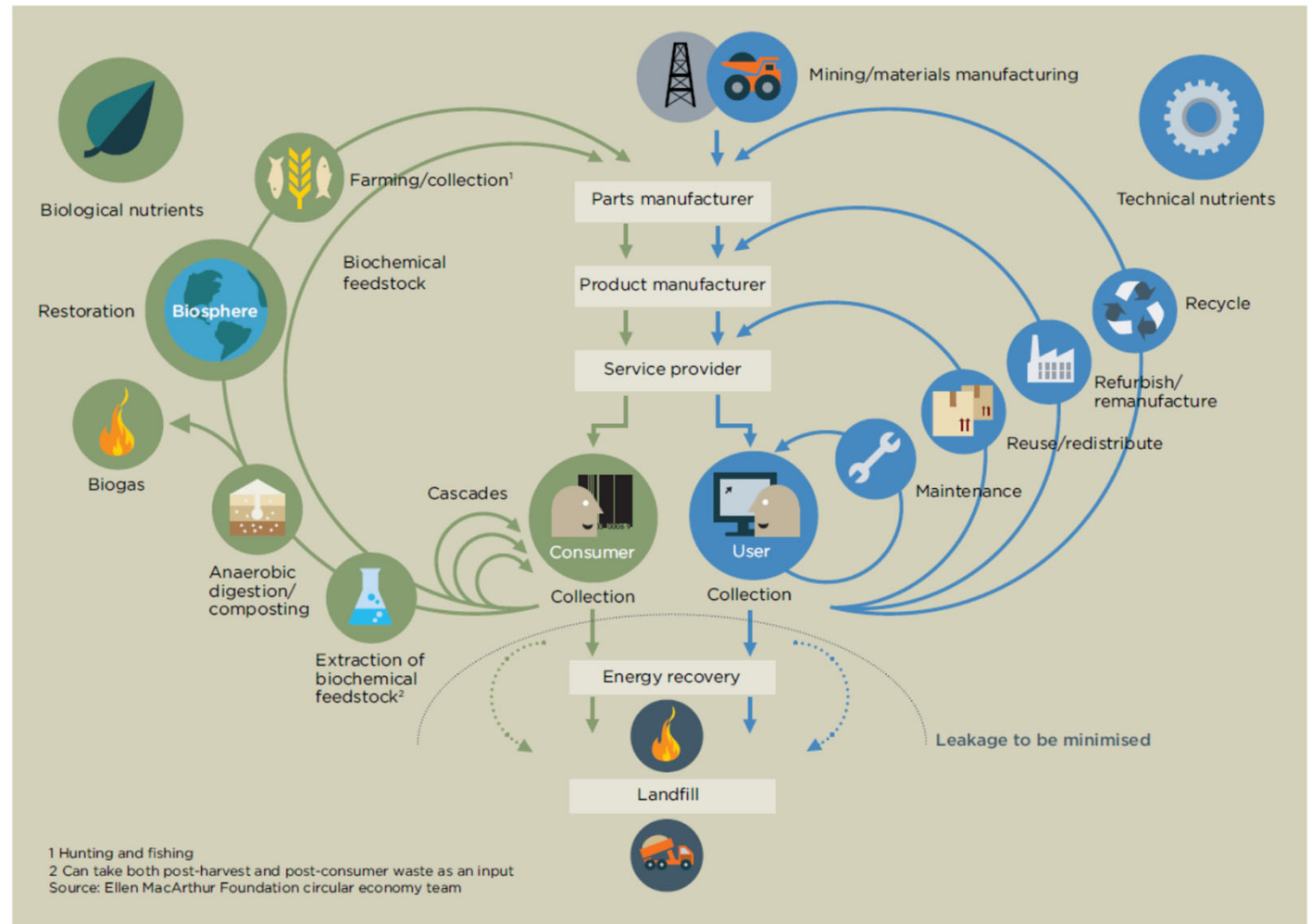


Circular Economy



Circular Economy

An industrial system that is restorative by design



Circular economy principles



**Design out waste and
pollution**



**Keep products and
materials in use**



**Regenerate natural
systems**

Moving towards circular business



Case studies and examples of circular economy in
action



Roetz-bikes



Fairphone

3 circular strategies (Atasu et al. 2021)

1. Retaining product ownership

- Circularity-as-a-service: Michelin Tyres, Philips Lighting/Signify, libraries.
- Negative cases: Interface Carpets pre-2000, Gorenje washing machines

2. Product life extension

- Long-lasting products: Patagonia, Miele.

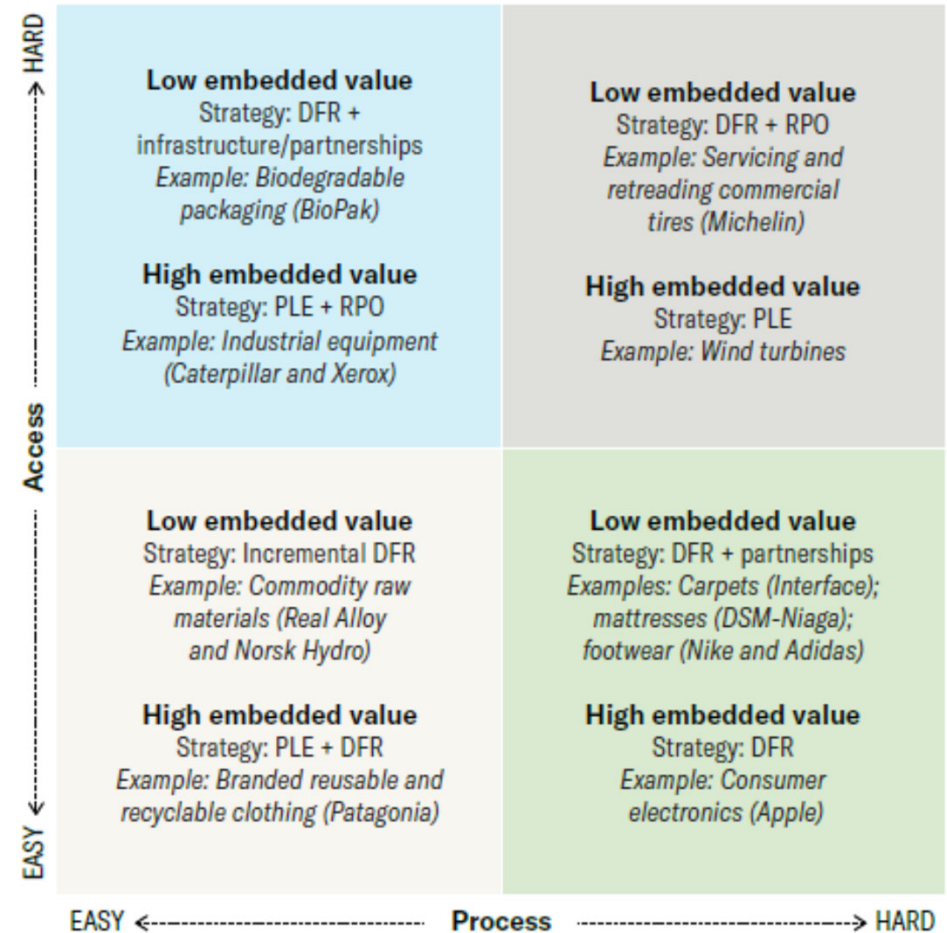
3. Design for recycling

- Eco-efficiency and innovation: Fairphone, Interface Carpets post-2000, Adidas.

Circular strategies (Atasu et al. 2021)

The right strategy depends on:

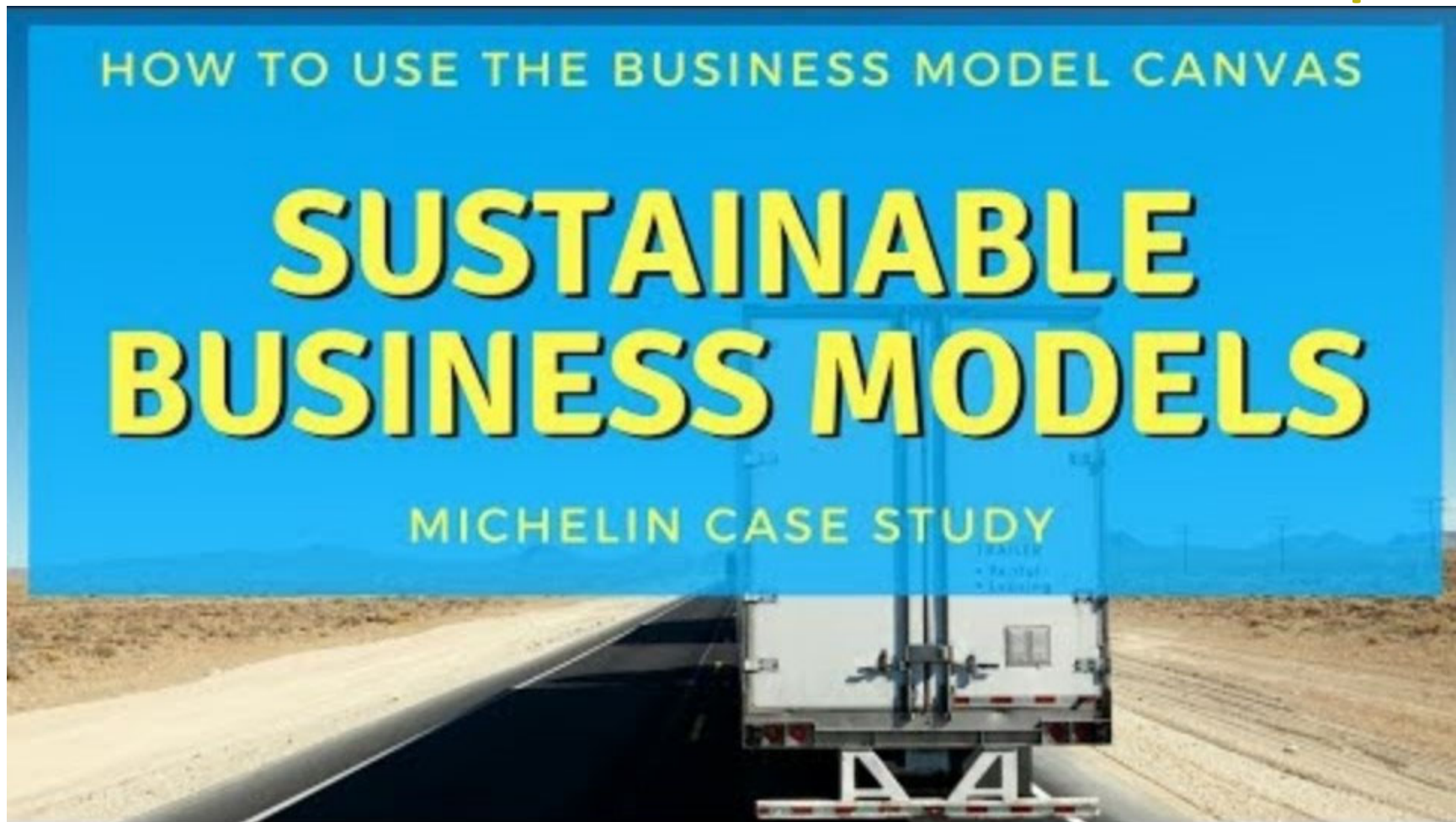
1. How easily can you **get the product back**?
2. How easily **value can be recovered** from it?



Source: Atasu, A., Dumas, C. & Van Wassenhove, L.N. 2021. The Circular Business Model. Harvard Business Review.



Sustainable Business Model Canvas: Michelin example



Systems change and different ways of thinking

1. Breakthrough thinking	2. Design thinking	3. Systems thinking
“an entrepreneur armed with a new piece of technology, ignores existing relationships, norms, and even laws to create a better product ”.	“Innovators study people in the contexts in which they will use an offering, empathize with them, divine their unarticulated needs , and then seek to reshape products and processes in ways that improve their experience ”.	“Systems thinking recognizes and embraces the complexity of organizational problems, rather than seeking to simplify them. It leads to innovations that make an entire system more sustainable and resilient , avoiding the side effects and collateral damage sometimes seen in the other two approaches.”
Examples: Google, Uber	Examples: Airbnb, Apple’s iPhone	Examples: Your next venture?

Four step framework toward systems change

→ To reap the advantages of systems thinking **you don't need to precisely model a complex adaptive system.**

1. **Define a desired future state:** Maple Leaf Foods went from meat packing to “the most sustainable protein company on Earth.”
2. **Reframe problems so that they resonate across stakeholders:** Farmers responded better to soil health than climate change as a frame.
3. **Focus on flows and relationships over discrete products:** Co-operators, a financial services and insurance company implemented “dry in place fans” for houses after floods, reducing landfill waste.
4. **Implement small nudges to gradually shift the system:** Stand association CSA implemented “ecology of actions” including education, materials standards, and pilot projects for the built environment.

Pace of change: After 200,000 years of humans...

- Widespread electrification, widespread use of cars, traffic light, widespread air travel, modern submarine, advances in materials, credit card, ATM, Barbie doll, modern management, frozen foods, microwave oven, canned beer, bubble gum, drones...
- Computer, internet, mobile phones, FM radio, color TV, video tape, cassette, CD/DVD, transistor, microchip, computer programming, online streaming, robot, GPS, virtual/augmented reality, inkjet/laser/3D printer, blockchain, cryptocurrency, digital assistants, robotic exoskeletons, Artificial Intelligence ...
- Penicillin/antibiotics, contraceptive pill, pacemaker, artificial hearts/organs, kidney dialysis, contact lenses, MRI, development of X-rays, test-tube baby, Viagra, vitamin supplements, chemotherapy, vaccines for various diseases, structure of DNA, gene splicing, human genome, stem cell research, genetically engineered animals...
- Nuclear power, atom bomb, fuel cell, jet engines, space travel...

Easter morning 1900: 5th Ave, New York City. Spot the automobile.



Source: US National Archives.

Source:
www.businessinsider.com/5th-ave-1900-vs-1913-2011-3?international=true&r=US&IR=T

**Easter morning 1913: 5th Ave, New York City.
Spot the horse.**



Source: George Grantham Bain Collection.

Source:
www.businessinsider.com/5th-ave-1900-vs-1913-2011-3?international=true&r=US&IR=T



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5. Recap and closing of general part

Picture source:

<https://www.cntraveler.com/gallery/where-to-see-the-best-fall-foliage-in-new-england>



Recap

1. Broader context of business

- Anthropocene and key sustainability trends
- Frameworks for sustainability

2. Business and climate change

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